

Instructor: Date:

Learner: Regional Branch:

Program: Center:

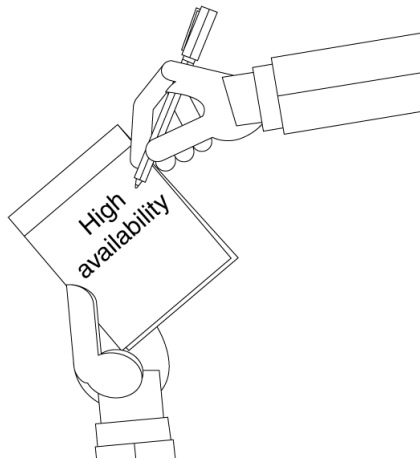
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Let's Start!

High Availability



Learning Outcome:

Exchange written and spoken information on principles, concepts, data base objects related to query analysis, and data base engines. Describe relational database, its concepts, and principles in spoken or written texts.

Can Do Statements:

- I can explain what high availability means.
- I can present a context to talk about high availability.
- I can go over the causes of downtime.

Ask your partner:

- How can a database provide a high availability access?
- How do databases prevent any adverse event if they are always online?





Let's Explore!

Scott and Lisa discuss how databases achieve* high availability

*"Achieve" is to accomplish.



Instructions: Read the text below.

Scott: Lisa, do you know what high availability is?

Lisa: Of course. It is the possibility of using any online system 24/7 from anywhere; that system is online even when the system presents problems.

Scott: So it is full-time available even with problems or failures happening around. But how can a system be kept online all the time?

Lisa: There are products and features to cover downtime, even if it is planned or not. For example, they can use Oracle's real application clusters, which allows Oracle database to run on a set of clustered servers; if one fails, the system will work with the remaining ones.

Scott: Wait a minute. I've heard of Oracle Data Guard, is it one of those programs for covering downtime?

Lisa: Yes. That one creates, manages and monitors databases that are on standby, so they can survive failures or data corruption. Those standby databases are basically consistent copies. Oracle Streams is another example, it sends and manages data transactions in a data stream, and you can decide what information is put into the stream. You can use Online Table Redefinition as well; it will reorganize scripts and tables, even while used by the client.

Scott: There are too many tools!

Lisa: Just understand that Oracle Database has many ways to protect the data. One of the last examples would be Automatic Store management. This one creates disk groups, which have a set of disks within to make grouping and information management easier. Finally, the Flash Recovery Area. It is a unified storage location for all recovery-related files that generates copies of files into disk groups.

Scott: It seems they are all related to one another!

Lisa: Indeed. That will allow the system to be available all the time: it's not called high availability just because.



**Let's Practice!**

There will be a new online system.

Instructions: Proofread the sentences. Correct the mistakes by writing the text on the lines provided.

From now on, our online system will had high availability. It means, you will being allowed to use it 24/7; as I know that you are travel next month, our system will is available to access from any place you visit. It willing be online even when there is any kind of problem because there will been products and features to cover downtime issues.

**Let's Practice!**

How does it run?

Instructions: Circle the correct answer to complete the sentence. Write it on the lines provided. Read the sentences aloud. Check your pronunciation with your instructor.

1. Oracle database _____ to run by using Oracle Real application clusters.
 - a. is allowed
 - b. are allow
 - c. allowed
 - d. is allow
2. The system will have products and features to avoid downtime even when it _____.
 - a. is not plan
 - b. is not planned
 - c. were a plan
 - d. are not planned
3. Oracle streams sends data transactions in a data stream, you just need to decide what information _____ into that stream.
 - a. are included
 - b. were included
 - c. is including
 - d. is included
4. You can also use Table redefinition to reorganize scripts and tables when it _____ by clients.
 - a. is used
 - b. are use
 - c. are used
 - d. is use
5. Data _____ by Oracle Database in many ways. For example, through Automatic Store management.
 - a. Protect
 - b. is protected
 - c. were protected
 - d. are protected





Extension Activity

Let's write about high availability!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for information about high availability.
- Identify high availability advantages.
- Write down a text explaining what will happen if there isn't high availability in a system.
- Share your text in the class and discuss about the topic.

Self-Assessment

Yes

No

- I search for information about high availability. ☐ Yes ☐ No
- I identify high availability advantages. ☐ Yes ☐ No
- I write a text explaining what will happen if there isn't high availability in a system. ☐ Yes ☐ No
- I share information about high availability and discuss about the topic. ☐ Yes ☐ No



Target Words

- **Availability** (noun): When a database is online and ready to be use anytime or anywhere.
e.g. Do you know what high availability is? Of course. When you use any online system and you can use it 24/7, from anywhere, and the system is online even when it presents problems.
- **Full-time** (adjective): That works or is present all the time.
e.g. it is full-time availability even with problems or failures happening around. But how can a system be kept online all the time?
- **Downtime** (noun): When a webpage or database is offline.
e.g. they have products and features to cover downtime, even if it is planned or not.
- **Standby** (noun): When data in a database is not being used.
e.g. That one creates, manages and monitors databases that are in standby, so they can survive failures or data corruption
- **Generate** (verb): To create or produce.
e.g. It generates copies of files into disk groups.
- **Just because** (idiom): For no specific reason
e.g. That will allow the system to be available all the time. It's not called high availability just because.



Learning Points

Future Simple Tense

We use the **future simple** to talk about events or actions that take place in the future.

You use **will** to create sentences in **future simple**.

We use **will** for:

1. Stating promises
2. Talking about actions in the long future
3. Make predictions of events.

Structure

Subject + will + verb + complement

Examples:

We **will** get more oracle features as soon as we get paid.

You **will** use oracle databases a lot in your new job.

Examples

- *Oracle database to run in a set of clustered servers, if one fails, the system will keep working with the remaining ones.*
- *You can use Online Table Redefinition as well; it will reorganize scripts and tables, even when it is used by the client.*

Causes of Downtime

There are many reasons why a database is in downtime.

Databases need to apply patches or upgrades, those are usually the reasons for scheduled downtimes, but sometimes the downtimes are due to reasons beyond their control, the following chart list the most common causes.

Can be scheduled by the administrators	Can never be controlled
Upgrades to software	Hardware failure
Upgrades to hardware	human error
Applying patches	Software failures
Servers Maintenance	Natural disasters

